ALI EDUCATIONAL

App features, updates and release review

Consolidated overview • 1 November 2026

Prepared for Musid and the shop owner. Covers the existing app, agreed changes, implementation instructions and required acceptance testing.

Important status distinction

You reported that the app has been built and many updates implemented. This report consolidates what we have specified and reviewed; it is not a certificate that every feature is implemented or that the current APK passed all tests. Screenshots show selected screens, not complete behavior. The latest app source, APK and end-to-end test results have not been independently reverified for this report.

Latest decisions take precedence

English interface and receipt headings; Urdu supported in product names. Receipt Preferences removed. Three current Pigmy schemes with future additions allowed. Deposits and interest-free withdrawals linked to Cash In/Out. Manual Google Drive / Files backup saving. Category creation and optional transaction photos. Preserve existing records through all upgrades.

Report contents

1. Product scope and design 2. Inventory and imports 3. Billing and receipts 4. Accounts and cash 5. Pigmy savings 6. Services 7. Reports and profit 8. Backup and recovery 9. Two-phone operation 10. Testing and distribution 11. Evidence, limitations and open decisions Appendix: detailed latest implementation addendum

1. Product scope and simple design

Ali Educational is an Android shop-management app for stationery and related shop goods. Stock, customer money owed, supplier dues and fast billing are central functions. Printing services and financial reporting extend the same records rather than creating disconnected systems.

The Owner phone is the main device. Core work is local and should remain usable offline, with no desktop required. A second phone may use the same app in a restricted Billing role. The design avoids separate billing implementations and unnecessary online logins.

The visual identity uses the supplied Ali logo and Ali Educational name. Dark charcoal/black with gold accents is the default direction; an optional light theme is retained. Large readable controls, simple labels and reachable actions matter more than decoration.

Settings uses a gear icon. Recent Activity is removed from Home, but underlying stock movements and audit history remain. The payment-screen quick buttons must no longer break words and amounts into vertical fragments. Forms must work with the keyboard open and larger system fonts.

The latest language decision removes the Urdu interface and Urdu receipt translation. App labels and receipts are English. Product names entered in Urdu or mixed scripts must remain intact in storage, search, bills, PDFs, imports, exports, sync and backups. Do not change the whole interface to RTL to display one product name.

Local fingerprint/device credential protection remains the preferred access model. Cold launch, resume and configured lock timing need testing. Pairing identifies a device role; it is not an online account or an unrestricted role-switch button.

2. Products, stock and spreadsheet entry

Current Add Stock & Product form

The supplied screenshots show Product Name, optional Barcode, Category, Brand, Pcs, Low Stock Alert, Base Cost per piece, Selling price per piece, Margin (%), Wholesale price per piece, Supplier Name, Bill No. and Transport Charges. The sample workbook was adapted to these fields. The precise Margin calculation must be checked in the app; the test sheet leaves it blank rather than assuming margin equals markup.

Products may have variants, categories/subcategories, optional photos and barcodes. Products without a barcode remain selectable through search, categories and favorites/recent products. Camera scanning uses existing product barcodes. No short-code system was requested.

Pieces and packs share underlying stock. Pack sizes can vary, and independent piece/pack prices are supported by the agreed plan. Wholesale per-piece price is not a pack price. Barcode lookup must respect the mapped unit and exclude stale or inappropriate archived mappings.

Editing a product name or selling price must not add its existing stock again. Receive additional stock, edit product details and correct a physical count are separate operations. Preserve history of quantity changes, damage, shop use and giveaways. Low-stock thresholds support defaults and product overrides.

Receiving and cost

Record received quantities, supplier, invoice details, purchase cost and delivery charges. Allocate delivery proportionally to purchase values with exact total paise preserved. Weighted-average costing is the agreed policy. For ₹500 stock plus ₹50 transport, total received inventory cost is ₹550. Historical sale costs must be preserved as snapshots.

Supplier balances must reflect the real liability; do not automatically assume every transport charge is payable to the supplier. Partial supplier payments and optional invoice/payment-proof attachments belong to the corresponding transaction history.

Excel import and validation

Use file selection, sheet choice, mapping, preview, validation and explicit confirmation. Support advertised XLS/XLSX/CSV formats and content providers. Reject corrupt, empty, wrong-type and inaccessible files with useful messages rather than crashes. Preserve leading-zero barcode text, exact currency, optional blanks and Urdu names.

Import must distinguish catalog updates, counted stock and additional receipts. Do not repeatedly add opening stock when the same file is retried. Detect duplicates, preserve batch history and avoid partial silent changes. Unsupported fields must be disclosed in preview, not mapped to unrelated columns.

The two original October XLS references were cash ledgers with Date, Notes, Description / Category, Income, Expense and Balance headings on row 5. They are not stock files. Product import should direct the user to Cash In/Out for those files.

The new synthetic workbook has 12 products and 372 pieces. Base cost totals ₹3,458; transport ₹117; landed stock cost ₹3,575; listed retail value ₹5,591; expected gross profit ₹2,016. Only Products is an import sheet. Instructions and Expected Results are checks, not import data. These figures are sample data, not real shop balances.

3. Billing, payments and receipts

Billing is now a major app function. Select by camera barcode, search, categories or favorites; adjust quantities and current-bill prices. Catalog price edits are separate from bill-only overrides. Hold and resume multiple drafts without deleting their only durable copy. Service-only and mixed bills should use the same checkout.

Whole-bill percentage or amount discounts remain distinct from unpaid customer credit. A ₹100 bill settled at ₹90 by discount is a ₹90 sale with ₹10 discount. If ₹10 is still owed, the sale remains ₹100 and requires a customer-linked balance. Do not create unassigned walk-in debt.

Cash Received must be directly editable for any valid rupee amount, including decimals. Exact sets the payable amount. Increment buttons and fixed-amount buttons must have unambiguous labels. A ₹48 bill with ₹50 cash returns ₹2 change; ₹47.50 requires explicit shortfall handling. Preserve tendered cash, applied payment and change separately.

Support the agreed cash/digital/credit and split-payment workflows with optional digital references. Saving twice must not duplicate a sale. Pack billing must use selected-unit prices correctly: one pack of ten at ₹100 is a ₹100 line and ten pieces out, not ₹1,000.

Returns, exchanges and cancellations need linked reversals, correct refund/credit choices and exactly-once stock effects. Later debt repayments must be considered when reversing a sale. Damage and saleable returns must not subtract sold stock a second time.

Latest receipt layout

Receipt Preferences is removed from Settings. Receipt actions belong on the saved-bill preview: WhatsApp/share, Save PDF and actual printing. Old automatic sharing/printing preferences must not trigger unexpectedly after upgrade.

Use a clean monochrome Ali logo on white, separate logo/title/subtitle spacing, aligned metadata, item columns and numeric totals. Wrap long descriptions and paginate long bills. Divider lines must not cross text. Show original receipt number and date from the selected persisted bill.

Remove the visible Not GST registered statement and any retain/keep-the-bill instruction. This is a presentation change, not permission to add GST, invent a GSTIN or label the document a GST tax invoice.

The latest reference receipt is AE-M-20261101-8261 dated 01 Nov 2026, 01:22 pm. Photocopying is 1 copy at ₹2; show ₹2 gross line amount, ₹2 subtotal, ₹1 discount (50%), ₹1 payable, ₹1 paid and ₹0 due. The supplied PDF used an ambiguous discounted line amount and had rules crossing the totals. A corrected visual reference and layout notes were created for Antigravity.

WhatsApp must receive a real valid PDF attachment with a readable content URI and permission. Preserve files long enough for the recipient app to read them. The user selects/confirms the recipient; opening WhatsApp does not prove delivery. A missing app or cancelled chooser must not change the saved bill. Do not send supplier costs, profit or private evidence with customer receipts.

4. Accounts, Cash In/Out and giveaways

Maintain customer and supplier names/details, outstanding balances and partial payments. Existing opening balances require explicit user input/import review. Customer repayments settle debt and are not new sales. Supplier payment is not automatically a new operating expense. Archive actions must not hide outstanding liabilities or erase history.

Cash In/Out needs explicit effective date and time on both directions, with calendar/time pickers and preserved values when editing. Store created/updated audit times separately. Backdated transactions must appear in the correct day/month and export. The exact policy for future-dated completed entries should be resolved rather than silently accepting them as current cash.

Categories and photo evidence

Each manual cash transaction has a dedicated Category selector separate from description/notes. Choose an existing category or create a new one without losing the form. Use stable IDs, duplicate-name validation and rename/archive rather than deleting referenced categories. Category/date filters and totals must reconcile.

Pigmy-linked deposits and withdrawals automatically use the Pigmy category and retain the individual scheme identity. A category is an organizational label, not the transaction accounting type. Selecting Pigmy manually must not fabricate a scheme transaction; guide the user to the linked workflow.

An optional photo can be captured or chosen from the gallery, previewed, replaced or removed. Copy it into durable managed storage; a temporary picker path is not permanent evidence. Failed processing must preserve the form and let the user explicitly retry or save without the photo. Editing evidence must not duplicate or change the financial entry.

Backup must include actual photo contents and relationship IDs. Restored photos must open on a new installation without access to the original phone path. Standalone Excel exports may show category and attachment presence but must not pretend to contain all photos. Owner-only financial evidence must not leak to restricted billing devices.

Cash-ledger spreadsheets

Preserve Notes when category is blank. Validate dates and currency; do not replace malformed dates with today. Skip titles, repeated headers, totals and closing-balance rows as transactions. Opening balance needs separate reviewed treatment. Warn about repeat/overlapping imports and ambiguous matches. Export the selected period only; an empty filter must not export everything.

Free products and donations

Products only; cash donations were not requested. Record product, quantity/unit, effective date, Free item or Donation reason, optional recipient/note and cost snapshot. Reduce stock once without sales income or debt. Show known giveaway costs separately from paid-sale COGS, and mark unknown cost. Reversal restores stock once and preserves history.

5. Pigmy savings: schemes, deposits and withdrawals

Expandable schemes

There are three schemes today, but the app must allow the Owner to add more. Each has a name, stable ID, usual deposit amount, separate history and totals. Show a combined summary and readable scrollable cards/list rather than squeezing all schemes horizontally. Rename/archive/reactivate while preserving historical records and links.

Replace the piggy-bank icon with a Bitcoin-style coin icon as requested. This is visual only: labels remain Pigmy savings and money stays in rupees. It introduces no cryptocurrency wallet, trading or conversions. Accessibility labels should describe Pigmy savings.

Deposits

Prefill the scheme's usual amount, allow changes and save only after confirming actual payment. Do not auto-post every day or mark all schemes paid when one is paid. Record date/time, funding source and optional note/reference. A cash deposit creates exactly one linked Cash Out savings transfer, atomically with the deposit.

Bank-funded deposits must not reduce physical cash. Edit/cancel updates or reverses the existing linked transfer once. Retry, restart and applicable sync must preserve exactly-once effects. The category is Pigmy, with a separate scheme name/ID. Savings deposits do not reduce operating profit as expenses.

Withdrawals without interest

The brother receives his deposited money back with no interest. Add Record withdrawal in each scheme, with amount, effective date/time, destination (shop cash or bank) and optional reference/note. A cash withdrawal creates a linked Cash In savings transfer; a bank receipt affects that destination, not physical cash. It is neither sales revenue nor profit.

Show Deposits, Withdrawals and Tracked balance. Balance is confirmed opening savings plus active deposits minus active withdrawals. Separate period movement from closing balance. Historical missing deposits must not be invented; any opening balance needs explicit amount/date confirmation and creates no new cash outflow.

For complete recorded savings of ₹10,000, withdrawing ₹3,000 leaves ₹7,000. Editing that withdrawal to ₹2,500 leaves ₹7,500; cancelling restores ₹10,000 and reverses the cash receipt once. An amount above tracked funds requires review of missing history/opening balance before posting, not fabricated balancing entries.

Preservation and acceptance

Old single-scheme data must not be copied into each new scheme. Preserve IDs and links; unknown attribution remains visible for Owner review. Backup/restore covers all schemes, deposits, withdrawals, openings, audit history and attachments. Add fourth/fifth schemes in testing and verify individual/combined totals, archive visibility and linked cash effects.

Synthetic test: deposits ₹100/₹200/₹300 yield ₹600 and three outflows. Edit the second to ₹250: ₹650. Cancel the third: ₹350. No change to sales/profit. Test withdrawals separately and confirm neither retry nor recovery replays cash movements.

6. Printing services and Owner expense reporting

Initial services are Printing, Photocopying and Sticker printing, plus Add service for reusable definitions. Printing options are A4 B&W and A4 Colour. Each new job requires its own entered rate; no default or last-used rates. Existing draft/held-line rates remain when editing that specific job.

The service definition stores identity, name and unit, not stock or a selling rate. Switching printing options clears an entered rate to avoid using a B&W price for colour. Quantity units must be explicit; sides, sheets, copies and stickers are not interchangeable. Photocopy/sticker charging units still require confirmation where not already resolved in implementation.

Services may share a bill with products. Only product lines deduct inventory; services do not automatically consume paper, ink or toner. Reversing a service does not restore product stock. Preserve service name, option, quantity, unit, entered rate and discount allocation as historical snapshots.

Example: products ₹160 and services ₹40 with 10% whole-bill discount produce ₹180 payable, ₹144 product revenue and ₹36 service revenue. Round line discount allocations to exactly the bill discount.

Only the Owner manages service expenses and accesses the service financial overview. Use one expense ledger with classification/target fields. Categories include paper, ink/toner, sticker sheets, repair/maintenance and other. Target a service or Shared services, with amount, date, note and optional evidence.

Show net service revenue, recorded service expenses and revenue minus recorded expenses. The difference is not exact service profit or available cash: supplies may span periods and sales can be unpaid. Do not allocate shared costs arbitrarily or count a reclassified expense twice.

Test credit repayments not becoming new service revenue, refunds not reducing revenue twice, void/edit history, no default rates, unauthorized Billing expense access, offline persistence, exports and backup. The earlier handoff includes 30 detailed service reporting/expense scenarios, all still applicable to implemented scope.

7. Sales & Profit and analysis

The recommended report entry point is Sales & Profit rather than an incomplete Gross Profit label. Keep product sales, cash collections, product COGS, product gross profit, operating expenses, services and savings transfers distinct.

Net product sales are itemized revenue after discounts/returns. Gross profit subtracts the corresponding saved cost snapshots. Missing costs must produce incomplete/known-cost labels rather than zero-cost assumptions and apparently exact 100% margins.

Expected stock profit is separate from actual profit realized through sales. It depends on current sellable quantities, selling prices and known landed costs. Negative stock or missing costs must not produce misleading projections. Weighted-average costing and proportional delivery allocation remain consistent across receiving and reports.

Daily summaries need explicit interpretation as additional unbilled sales or a separate summary. Never count a full-day total plus all its underlying bills twice. Later customer repayments are collections, not additional revenue. Supplier settlements, Pigmy transfers and inventory purchases are not all ordinary operating expenses.

Use consistent Today/Week/Month/custom ranges and daily/weekly/monthly graph grouping. Apply identical filters to cards, graphs, drill-downs, exports and incomplete-data warnings. Giveaways show separate known costs. Service comparison remains labelled as revenue minus recorded expenses, not exact net profit.

When the second phone has pending or unknown sync state, clearly indicate report incompleteness. A disconnected device's pending count cannot honestly be displayed as confirmed zero. Test boundary dates, backdated entries, changed catalog prices/costs, reversals, unknown cost and exported reconciliation.

8. Backup Now, Google Drive and recovery

Backup Now previously had a user-reported nonworking action. Verify the full button-to-service-to-file-to-result path. Show progress, avoid concurrent jobs, refresh the backup list and update last successful backup only after real validation. Failures need actionable errors and retry; do not leave a permanent spinner or false success.

Back up a consistent database snapshot including committed WAL data, not a casually copied live DB. Include products, stock history, bills/lines, accounts/payments, expenses/cash, services, Pigmy, categories, held bills, relevant settings and actual attachments/branding. Use a manifest with true format/schema/version details and checks, not decorative Verified labels.

The observed backup screenshot showed ali_educational.db and manifest.json. Those filenames and a small file size are not proof of completeness or failure. Records can be compact, but photos/settings may require extra contents. The current archive contents must be inspected and restored to establish what is actually covered.

Confirmed Drive policy

Keep manual on-demand Save Copy to Google Drive / Files. Create/validate a local backup, then let the user select an available provider/account/folder using Android's native picker. No unattended OAuth-based background upload work for this release.

Distinguish local backup creation from successful provider write and verified cloud delivery. Drive may queue an upload while offline. A picker/share launch alone proves neither saving nor upload. Show available filename/destination information and separate last-local-backup and last-external-save states. Provide a simple reminder to save an external copy, especially before updates.

Keep dated copies and retain the valid local archive when external saving fails. Handle cancellation, missing Drive, revoked permission, storage limits, offline behavior and interrupted writes. Do not silently delete older backups or invent cloud paths.

Restore safety

Validate archive structure, schema compatibility, integrity and attachment coverage in staging before touching live data. Reject unsafe paths, missing critical contents and unsupported versions. Require confirmation, preserve a consistent pre-restore recovery copy and reopen the database safely. Failures must leave prior data recoverable.

Sync and backup differ. Owner backup includes accepted bills, not unknown unsent bills on the other phone. Restoring requires reconciliation of event IDs/acknowledgements, not blind replay or overwriting another device role. Sensitive authentication/signing material is not ordinary backup content.

9. Two phones, one app

Use one shared Android app and compatible signed APK with Owner and Billing roles. Both phones are on the same shop Wi-Fi; no desktop server or remote-location sync is part of this phase. Optional capability stays off until explicitly paired and tested. Owner workflows must continue when the second phone is absent.

Pair through an Owner-approved short-lived invitation and authenticated encrypted connection. Send only role-appropriate catalog/data, not supplier costs, service expenses or full Owner backups. Billing must not promote itself through an unprotected selector.

Billing may override current-bill prices and whole-bill discounts but cannot change saved catalog prices. Permission for customer creation, credit, repayments, returns/cancellations, reusable-service creation and visibility of other bills remains to be confirmed where not resolved later.

Offline billing saves completed bills and outbound events atomically, keeps durable pending history and delivers on reconnection. Use stable unique event/transaction IDs; retries may repeat messages but must apply stock/money effects once. Acknowledgements occur after commit. Updated stock must not deduct an already acknowledged local sale again.

Preserve original timestamps, prices, units and receipt numbers. Show last-known stock while offline and retain both real sales if the last item was sold on each device; flag shortage rather than discarding a bill. Handle changed/archived products and revoked devices with visible review states.

Delayed offline cost assignment needs a documented policy, not an unsupported exact historical-profit claim. Invoice series must avoid offline collisions and survive backup/replacement. Test interruption at local save, remote commit and acknowledgement boundaries, plus screen-off, battery restrictions, changing addresses and device restart.

Two real phones are required for final connectivity/lifecycle evidence. If only one is available, mark two-phone hardware behavior unverified instead of claiming full completion.

10. Final testing and release handoff

Test fixtures and independent checks

Use synthetic spreadsheets in a separate app/database. The latest generated workbook is for adding products/stock. Test valid data, bad files, duplicates, malformed dates/currency, leading-zero identifiers and optional fields. Do not silently import test records into the brother's real catalog.

Perform actual transactions after import: sales, discounts, credit repayments, supplier payments, services, Pigmy deposits/withdrawals, giveaways and photos. Record an expected-value manifest with IDs, quantities, amounts and relations. Create a backup via the real UI, save/retrieve externally and restore into another disposable installation. Compare contents, not just record counts or ZIP size.

Rehearse migration from supported old schemas without destructive fallback. Test storage exhaustion, interruption, failed restore, missing attachments and corrupt archives. Protect the current database and validate recoverability before proposing a live update.

Evidence and acceptance gate

Test actual release-equivalent/minified Android behavior for PDF fonts/logo, file pickers, content URI sharing, camera, lock and backup. JVM tests are useful but cannot prove platform compatibility. Preserve expected/actual outcomes, build/device details, screenshots/documents and pass/fail/unverified status per scenario.

Do not release with critical crashes, data loss, unexplained stock/money discrepancies or failed recovery/migration. Rerun affected checks after fixes. A completed checklist or successful build is not feature evidence. Review exact artifact identity and any differences from the tested variant.

Build and share

Produce one signed release APK, compatible with the existing application ID/signing key and incremented version code. Exclude private reference data, test records, debug tools and secrets. Provide filename, version, size, SHA-256, release notes, setup/pairing instructions, test results and limitations.

Ask explicit approval before updating the production device. Never uninstall or clear data to bypass a signature/migration problem. Older APK installation is not a safe automatic rollback after a database upgrade; recovery must account for records created since the backup.

The APK can be downloaded locally and shared as a file using an appropriate channel, including the user's intended WhatsApp document flow where supported. A local workspace link is not a public link for the brother. Do not fabricate a hosted download URL or upload publicly without authorization. Send the APK only, not signing keys/passwords. After an approved update, use nondestructive smoke checks.

11. Evidence and remaining decisions

What this conversation actually produced

Detailed Markdown handoffs, the combined two-phone/service specification, repair/test plans, the corrected receipt reference image and layout notes, and a synthetic product-stock XLSX workbook. Earlier isolated source review and limited JVM tests identified defects; those historical checks are not evidence that the later Antigravity-built app is fixed.

The latest supplied screenshots show Home/Pigmy and payment/stock-entry layouts. The user confirmed Pigmy Daily opened nothing at that time and reported receipt/backup issues. Later user reports indicate implementations progressed, but this PDF does not independently certify the current build.

The AE-M-20261101-8261 PDF was rendered and inspected. Logo spacing and intersecting rules were observed, and the discounted item amount was ambiguous relative to subtotal. A visual layout reference was generated; it is not a production receipt renderer or proof that WhatsApp output was repaired.

The stock workbook totals were calculated and its expected-results rendering checked. It has not been imported into the latest app by this report-generation task. Current importer support for supplier, wholesale and transport columns still needs preview/device verification.

Questions that must not be silently invented

Confirm unresolved Billing-role permissions, exact service units, receipt series, delayed offline-sale costing and any future-dated cash policy. Confirm historical opening savings before creating an opening balance. Existing source may resolve some choices; inspect before asking the user again.

Optional local AI was discussed earlier but remains lower priority and outside the current release work: read-only answers/explanations, optional activation and no external shop-data transmission. No current AI implementation or quality is claimed.

How to use this report

Use Sections 1–10 as a consolidated feature/update review. The appendix retains the detailed latest addendum so Antigravity has exact category/photo, Pigmy withdrawal, manual Drive and receipt/icon requirements. Latest decisions override older deposits-only, fixed-three-schemes or bilingual-interface wording retained in historical specifications. Request build-specific evidence before treating any requirement as completed.

Appendix — Detailed latest implementation addendum

The following text is retained from ANTIGRAVITY_LATEST_CHANGES_AND_BACKUP_DECISION.md. It is an implementation/test specification, not a verified completion report. Later override sections take precedence over older scope wording.

Ali Educational — Latest changes and answer to Antigravity

Date: 1 November 2026

Latest approved Cash In/Out addition — categories and photo evidence

The user confirmed that each Cash In/Out transaction needs a dedicated Category selector, with selection from existing categories or creation of a new category, plus an optional photo attachment.

Entry and detail screens

- Show Category as a clearly labelled field separate from description/notes, amount, date/time and payment source/destination.

- Provide a searchable existing-category list and an Add category action. Creating a category from an entry form returns to the same form with the new category selected, preserving all entered values and any photo.

- Require a category for new manually entered transactions, with General available when appropriate. Existing uncategorized records remain visible and must not disappear from totals.

- Preserve existing user categories. Suggested familiar examples include Rent, Electricity and Transport; examples are not instructions to overwrite current classifications.

- Category creation needs a nonblank trimmed name; prevent confusing duplicates using case/whitespace-normalized matching. Use stable category IDs so renaming does not break records.

- Allow Owner rename/archive. Archive hides a category from new selection but preserves historical labels, records and reporting; do not cascade-delete transactions.

- Show category and an attachment indicator in history/detail. Add category filtering with an All categories option; date/source filters and totals must continue to reconcile.

- Categories are organizational labels, separate from accounting type. Selecting a category must not silently convert a savings transfer into an operating expense or sale.

Pigmy integration

- Automatically assign the Pigmy category to linked deposits and withdrawals. Keep scheme ID/name and deposit/withdrawal ID as separate data; do not encode all logic only in the text label.

- Deposit from cash = linked Cash Out savings transfer; withdrawal into cash = linked Cash In savings transfer. Neither is sales income or an operating expense.

- Protect the system category's identity even if its display name changes. Manual classification as Pigmy does not automatically create a scheme deposit or withdrawal; guide the user to the Pigmy workflow when a linked scheme transaction is intended.

- For linked entries, perform financial edits/cancellation through the corresponding Pigmy record or a shared atomic command so the two records cannot diverge. Evidence/category display must not produce another cash entry.

Optional photo attachment

- Offer Attach photo with Take photo and Choose photo using supported Android camera/photo-picker flows. One optional photo per transaction is sufficient for this requested scope; do not add unrelated document-management features.

- Show a thumbnail and allow full preview, replacement or removal before saving. Allow the Owner to add/replace/remove evidence on an existing entry without changing its amount/date or creating a duplicate transaction.

- Cancellation or denied permission leaves the entered form intact. Show actionable errors for unreadable/unsupported images or insufficient storage. Do not silently drop the photo while reporting successful attachment.

- Copy the selected image into durable app-managed attachment storage, retaining correct orientation and enough resolution to read receipt details. A temporary picker URI or camera cache path is not permanent evidence storage.

- Coordinate file and database persistence so interrupted saves cannot leave a transaction claiming a nonexistent attachment. Preserve the previous photo until replacement succeeds. Clean up abandoned temporary files safely.

- If image processing fails, let the user explicitly retry or choose Save without photo. Saving without evidence must not happen silently.

- Optional photo/note editing on Pigmy-linked entries must preserve the same financial record and cash link. Use a shared attachment reference or a single clearly defined owner record to avoid inconsistent duplicate copies.

- Respect existing Owner/Billing restrictions. Do not send Owner-only financial evidence to a restricted Billing phone. Do not attach transaction evidence to customer WhatsApp receipts automatically.

Backup, restore and exports

- Include actual photo bytes, attachment IDs, transaction relationships and required metadata in complete backups, including manual copies saved through Drive/Files. Backing up only a phone-local path is insufficient.

- Restore attachments into valid new locations and reconnect references. Verify a photo can be opened after restore on a separate installation, even when the original phone path is unavailable.

- Preserve category definitions, archived status and stable relationships through migration, backup, restore and applicable sync.

- Add a Category column to cash spreadsheet exports and accept it when importing supported layouts. Keep Notes separate. Older files without a category must show a clear preview/default rather than being rejected or silently misclassified.

- Spreadsheet exports can show an Attachment present indicator; clearly state that a standalone spreadsheet does not contain the original photo or constitute a complete backup. Do not export private temporary paths as useful attachment links.

Acceptance tests

1. Create a cash entry, select a category, save/restart and verify category, amount, date/time and notes remain correct.

2. Add a new category mid-form: form values/photo survive and the new category becomes selected. Reject blank/duplicate names without duplicate records.

3. Rename/archive a used category: historical transactions remain accessible and counted; new-entry selection respects archive state.

4. Filter by category and date: matching transaction totals reconcile with All categories and exports.

5. Record Pigmy deposits and withdrawals in several schemes: automatic category, correct scheme attribution, one cash effect and no expense/profit misclassification.

6. Add a photo from camera and gallery; verify orientation, readability, restart persistence and full preview. Cancel/deny permission without losing entered values.

7. Replace/remove a photo on an existing entry: no change to cash totals, no duplicate transaction, old image retained until replacement commits.

8. Simulate storage failure/interruption during save or replacement: recoverable form/error and consistent transaction/attachment references.

9. Back up categorized entries with photos, save through Drive/Files, retrieve and restore in isolation. Compare categories, IDs, amounts, timestamps and image contents; open restored photos successfully.

10. Import old/new cash-ledger layouts and re-export: categories/notes remain distinct; spreadsheet attachment indicator does not falsely promise photo recovery.

11. Test repeated taps, linked-record editing and existing sync retries for exactly-once financial effects; verify restricted-role access to entries/photos.

This is an implementation and test specification only. No app changes or tests are performed by this documentation update.

Latest approved Pigmy expansion — withdrawals without interest

This section supersedes the earlier deposits-only/no-withdrawals restriction. The user confirmed that his brother withdraws deposited savings and receives NO interest. Preserve expandable schemes and all earlier deposit/cash-link behavior.

Withdrawal flow

- Add Record withdrawal to each scheme's detail screen. Preselect the scheme; enter a positive amount, effective date/time, destination (shop cash or bank) and optional note/reference. Save only after confirmation of actual receipt.

- A withdrawal received into shop cash automatically creates exactly one linked Cash In entry labelled “Pigmy withdrawal — [scheme name]”. A bank receipt must not increase physical shop cash.

- Classify it as return of savings/transfer, not sales revenue, service revenue, customer repayment or profit. Do not add interest fields, calculated interest or interest income.

- Save withdrawal and destination-ledger effect atomically using stable IDs. Repeated taps, retries and sync must not duplicate effects.

- Editing/cancelling updates or reverses the linked receipt once; retain audit history. Changing destination reverses the old destination effect and applies the new one atomically.

Scheme balances and history

- Show Deposits, Withdrawals and Tracked balance separately for each scheme and in the combined summary, applying date ranges consistently.

- Balance = confirmed opening balance + active deposits − active withdrawals. Distinguish period net movement from closing balance; use earlier activity when calculating an ending balance.

- Keep a dated combined history with direction and destination/source, plus separate filters for deposits and withdrawals. Cancellation status remains visible in history but excluded from active totals.

- Label this a tracked balance, not a bank-verified figure. If historical deposits are missing, clearly mark incomplete history; do not invent deposits or silently treat an unknown opening balance as confirmed zero.

- Before introducing a historical opening balance, ask the Owner to confirm the amount and effective date. This establishes starting savings only and must NOT create a new Cash Out entry or be counted as a new deposit.

- If withdrawal exceeds available tracked savings, show the discrepancy and require review/correction of missing history or opening balance before posting. Do not silently create negative savings or fabricate a balancing deposit.

- Preserve current deposit/transfer IDs through migration. Include withdrawals, opening-balance provenance, links and audit history in backup/restore, exports and relevant sync. Owner-only administration remains required.

Required regression tests

1. Complete test history: deposits ₹10,000, withdrawal ₹3,000 to shop cash → tracked savings ₹7,000 and one Cash In transfer ₹3,000; sales and profit unchanged.

2. Same withdrawal to bank → savings ₹7,000 and bank receipt ₹3,000; physical shop cash unchanged.

3. Edit ₹3,000 withdrawal to ₹2,500 → balance ₹7,500 and linked receipt ₹2,500, not an additional receipt. Cancel → balance ₹10,000 with receipt reversed once.

4. Double tap/retry/restart/sync: one committed withdrawal and one destination effect. Cancelling an unsaved form changes nothing.

5. Validate missing scheme/destination, blank/zero/negative amount and withdrawal above tracked funds. Explain incomplete history without guessing a starting amount.

6. Confirm a historical opening balance: savings changes appropriately, no new cash transfer or deposit income is created. Date ranges distinguish opening, movement and closing totals correctly.

7. Use several schemes: withdrawing from A leaves B/C unchanged; combined totals reconcile. Renaming a scheme retains links.

8. Restore an isolated backup and compare deposits, withdrawals, opening values, effective timestamps, balances and linked cash records. No replayed cash entries.

9. Reports/exports show withdrawals as savings transfers and contain no interest income. Existing cash filters, deposits and bill collections remain correct.

These are implementation requirements and future tests for Antigravity, not confirmation that the app already supports withdrawals.

Message to Antigravity

Implement the latest requirements below in the current app. Preserve all completed work and existing shop data. This document specifies requested changes and tests; it does not claim the app has already been changed.

My answer to your backup-policy question is: **keep manual on-demand “Save Copy to Google Drive / Files” for this release. Do not implement unattended automatic background uploads or require Google Cloud OAuth credentials for this phase.**

Follow these detailed requirements and complete isolated testing before asking to install a production update. Do not repeat unrelated earlier implementation work.

1. Manual Google Drive / Files backup — confirmed decision

User flow

1. The user taps Back up now. Create and validate a consistent, complete local backup first.

2. Show the result and offer Save Copy to Google Drive / Files.

3. Open Android's native document picker so the user can select an available Drive account/folder or local destination.

4. Write the complete archive through the selected content URI and close the stream successfully. Preserve the original local backup.

5. Display the filename and selected destination information that the provider actually exposes. Do not fabricate a filesystem path for a cloud document.

Scope and honest status

- Keep the existing native picker approach; no custom Google login/password collection or OAuth project setup for this release.

- Drive must be available through a supported installed document provider. If unavailable, explain this and allow another destination or cancellation without losing the local backup.

- Distinguish “Local backup created” from “Copy saved to selected provider”. A provider can accept a file locally while its cloud upload remains pending.

- Do not say “Uploaded to Google Drive” merely because the picker opened or a share intent launched. Show cloud-confirmed status only when the implementation can actually verify it.

- Show last successful local backup separately from last completed external save. Preserve previous successful timestamps when a new attempt fails.

- Add a simple visible reminder to save an external copy after shop closing and before updates. Do not claim a reminder performs an upload. Notification scheduling is not an approved automatic-upload feature.

- Use dated filenames and retain previous backups. Do not silently overwrite or delete old files.

- User cancellation, offline providers, revoked access, full storage, unavailable account and write failures must leave a clear recoverable state and allow retry.

- Preserve records, settings, attachments and branding in backup coverage. A local database file and metadata alone do not prove everything is covered.

- Drive is a backup destination, not a live database or substitute for two-phone synchronization. Show when pending billing-phone records have not reached the Owner backup.

Required Drive/backup verification

Create synthetic Excel stock and cash-ledger files, import into a separate test app, perform representative transactions and record expected quantities/balances/IDs. Create a backup through the real button, save a copy to a test Drive folder, retrieve it and restore into another disposable database. Compare all records, links, amounts, dates, settings, branding and attachments against the expected manifest.

Test cancellation, unavailable Drive, offline save behavior, write failure, retries, two dated backups, corrupted archive and unsupported schema. Show actual restore evidence, not just a successful picker or nonzero ZIP size. Never restore over live shop records or send test receipts to real customers.

2. Pigmy accounts and automatic Cash In/Out linkage

The brother currently has THREE schemes, with the ability to add more later. Each scheme has its own name, usual daily deposit amount, history and totals; show a combined recorded-deposits total too. Do not hardcode real names/amounts or impose a three-scheme limit.

- Add deposit preselects the chosen scheme and prefills its usual amount. The user can change it and confirms actual payment before saving. No automatic daily postings.

- Save stable scheme/deposit/transfer IDs. A confirmed cash deposit creates exactly one linked Cash Out savings transfer labelled with the scheme name. Save both atomically.

- A bank-funded deposit must not reduce physical cash. Record the actual funding source clearly.

- Editing/cancelling a deposit updates/reverses its existing linked transfer once. Changing its scheme changes attribution without creating another outflow.

- Savings transfers do not count as operating expenses or reduce product profit. No duplicate manual Cash Out entry should be needed.

- Deposits only: no interest, withdrawals or maturity payouts in this phase. Label totals Recorded deposits rather than claiming a verified bank balance.

- Preserve existing deposits when migrating: do not copy them into every scheme. Keep unknown attribution visible for Owner review without generating another cash effect.

- Restrict scheme administration to Owner, preserve history on archive and include schemes/links in backup, restore and applicable sync.

Tests: create schemes A/B/C with synthetic deposits ₹100/₹200/₹300 → totals ₹100/₹200/₹300 and combined ₹600, with exactly three cash transfers. Edit B to ₹250 → combined ₹650. Cancel C → active combined ₹350. Verify no profit effect and no duplicate transfers after retries/restart. Add D and E, rename/archive/reactivate schemes, restore all five and compare IDs, totals and links. These are test values, not production defaults.

3. Receipt cleanup and scheme icon changes

Latest overrides — receipt cleanup, expandable schemes and icon

These instructions supersede conflicting earlier requirements below. Implement in the current app without undoing completed fixes.

Receipt placement and removal of unwanted text

Latest user-supplied receipt reference:

The user reports that placement is STILL incorrect. This documentation update does not establish the precise layout defect from an independent PDF inspection. Antigravity must render and inspect this actual PDF, identify the remaining alignment/spacing issues and compare corrected generated output before declaring completion.

- Remove the visible “Not GST registered” statement from the receipt header/body wherever generated. This is a presentation change only: do not add GST charges, GSTIN or a GST Tax Invoice label, and do not change the shop's underlying tax status.

- Remove the footer asking the customer to retain/keep the bill, including equivalent wording such as “Please retain this bill”. Do not replace it with another retention instruction.

- Remove the empty layout space left by these lines so the header/body/footer remain compact and balanced.

- Correct the actual logo/shop-heading alignment, receipt metadata, item columns, totals/payment placement and margins after inspecting the PDF. Preserve all necessary bill information and exact amounts.

- Use measured layout constraints and text wrapping instead of guessed absolute positions that collide on longer content. Verify short, long and multi-page receipts, English labels and mixed/Urdu product names.

- Apply the cleanup consistently to receipt preview, PDF, WhatsApp attachment, saved copy and printing. Regenerate from the selected saved bill; do not alter historical transaction data.

- Deliver before/after rendered evidence from the supplied example with a short explanation of the specific placement corrections. A successful share alone does not prove the layout is fixed.

Tests: assert both unwanted statements are absent in generated receipt text; visually inspect no blank reserved bands/overlaps; verify logo proportions and aligned monetary columns; compare item/payment totals with stored records; test long names, multiple pages, mixed services/products and a fresh WhatsApp attachment rather than a cached old PDF.

Expandable pigmy schemes

There are THREE schemes today, but three is NOT a permanent app limit. Replace any fixed-three storage, UI or validation assumptions with a data-driven scheme list.

- Owner can use Add scheme to create more schemes in future, entering a name and usual deposit amount. No real scheme names or amounts may be fabricated.

- Each new scheme gets a stable unique ID and the same separate history, totals, add-deposit action, date filters and linked Cash In/Out behavior as existing schemes.

- Validate required name and positive usual amount; warn about duplicate names without confusing scheme identity. Keep names editable and preserve references when renamed.

- Keep the combined summary at the top and use a scrollable list of readable cards/rows; do not cram all schemes into a single horizontal row.

- Provide archive/reactivate for schemes with history instead of destructive deletion. Archived schemes remain accessible in historical reports and included in all-scheme historical totals for the selected period, but do not accept new deposits until reactivated. Clearly label active-only filters if offered.

- Migration keeps the current three schemes and every deposit/link intact. Adding a fourth scheme must not reset the first three or create any payment automatically.

- Backup/restore, exports, permissions and applicable sync must support any recorded number of schemes without hardcoded indexes or exactly-three validation.

Tests: add a fourth and fifth scheme; deposit in each and reconcile individual/combined totals and cash links; rename/archive/reactivate; verify archived history remains counted correctly; restore five schemes with all IDs/links; test long names and enough schemes to require scrolling; duplicate taps must not create unintended duplicates.

Replace piggy icon with Bitcoin-style icon

Replace the piggy/piggy-bank icon used for Pigmy Daily and its scheme screens with a clean Bitcoin-style circular coin icon containing the recognizable B symbol with vertical strokes, as requested.

- This is a visual icon change only. Keep labels “Pigmy” or “Pigmy savings” and all amounts in rupees. Do not rename the feature Bitcoin, convert deposits to cryptocurrency, add wallets/trading or introduce crypto calculations.

- Use a locally bundled vector asset or an appropriate existing icon asset, with gold/charcoal/white colors consistent with the current theme and readable contrast in light/dark modes.

- Apply consistently to the Home shortcut and relevant Pigmy screens. Accessibility descriptions must describe “Pigmy savings”, not a Bitcoin account.

- Verify sharp rendering, adequate touch-target spacing, no missing asset in release/minified builds and no remaining piggy-bank icon in this feature.

4. Before asking me to install the update

- Identify the current project/build and preserve existing work. Use an isolated test app/database.

- Run the tests above and regressions for affected billing, reports, receipts, navigation, cash entries and backup workflows. Distinguish executed tests from unverified items.

- Show actual before/after rendered receipt evidence, phone screenshots, scheme/cash reconciliation results and successful isolated backup restoration.

- Verify nondestructive migration and compatible signing identity. Never uninstall the live app, clear its storage or use destructive migration to make the update install.

- Provide candidate APK version/code/hash, passed/failed/unverified tests, limitations and recovery steps.

- Request explicit approval before updating the production app. A connected phone is not authorization to overwrite its live data.

Keep the app interface and receipt headings English only. Urdu/mixed-script product names must still render correctly. Receipt Preferences remains removed from Settings. Do not introduce unrelated new features or reset earlier completed work.